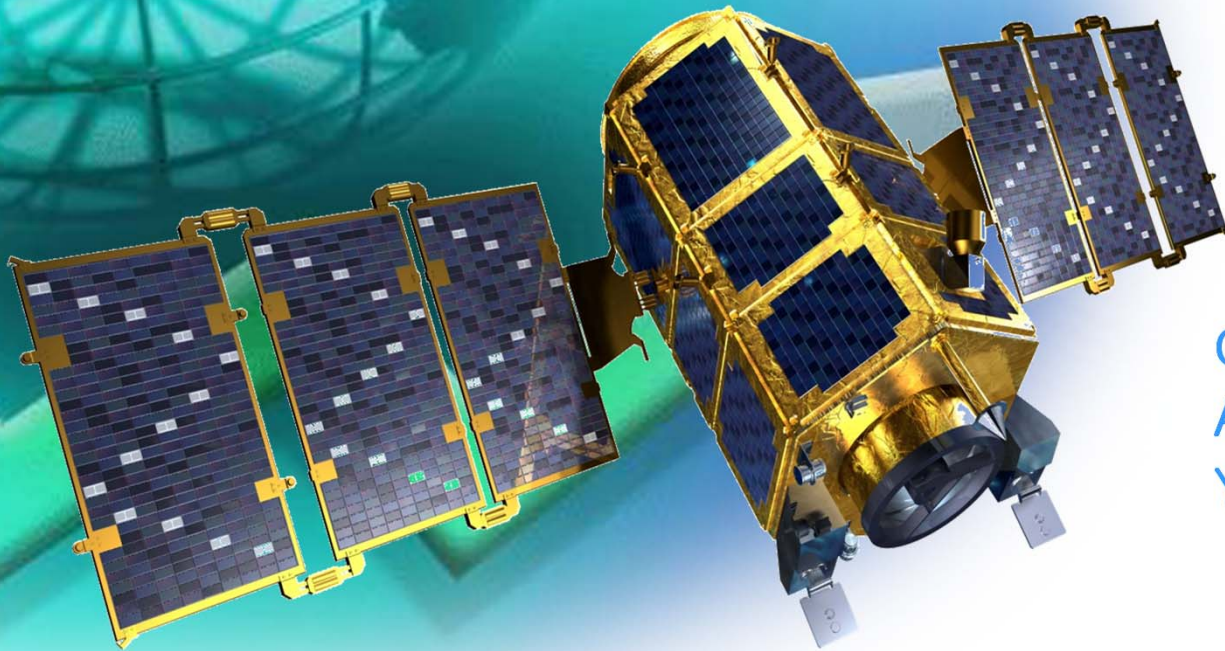
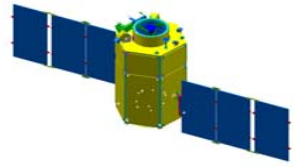


Fall 2011

Spacecraft Systems



Chandeok Park
Astrodynamics & Control Lab.
Yonsei University



Chapter 20

COST MODELING

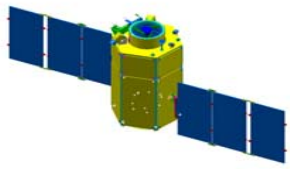
20.1 Introduction to Cost Analysis

20.2 The Parametric Cost Estimation Process

20.3 Cost Estimation Relationship

20.4 Other Topics

20.5 FireSat Example



20.1 Introduction to Cost Analysis

◆ Cost is an engineering parameter that varies with:

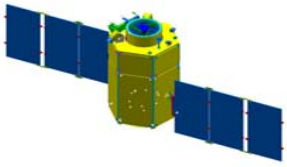
- Physical Parameters
- Technology
- Management Methods

◆ System's Cost depends on:

- Size
- Complexity
- Technological Innovation
- Design Life
- Schedule

◆ System's Cost is also a function of:

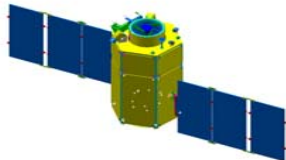
- Risk Tolerance
- Methods for Reducing Risk
- Management Style
- Documentation Requirements
- Project-Managements Controls
- Size of the Performing Organizations



20.1 Intro. to Cost Analysis

◆ Life–Cycle Costs

- Total Mission Cost from planning through end–of–life
- (RDT&E) Phase: Research, Development, Test, & Evaluation includes
 - Design
 - Analysis
 - Test of Breadboards, Prototypes and Qualification units
- Production Phase
 - Incorporates the cost of producing flight units and launching them
 - Theoretical First Unit (TFU)
 - A definition used to model costs
 - Represents the first flight–qualified satellite off the line
 - For single satellite missions, the TFU is the flight article
 - For multiple units, production cost is estimated using a learning curve factor applied to the TFU cost
- (O&M) phase: Operations & Maintenance consists of
 - Ongoing operations costs
 - Maintenance costs
 - Includes S/C unit replacements and software maintenance



20.1 Intro. to Cost Analysis

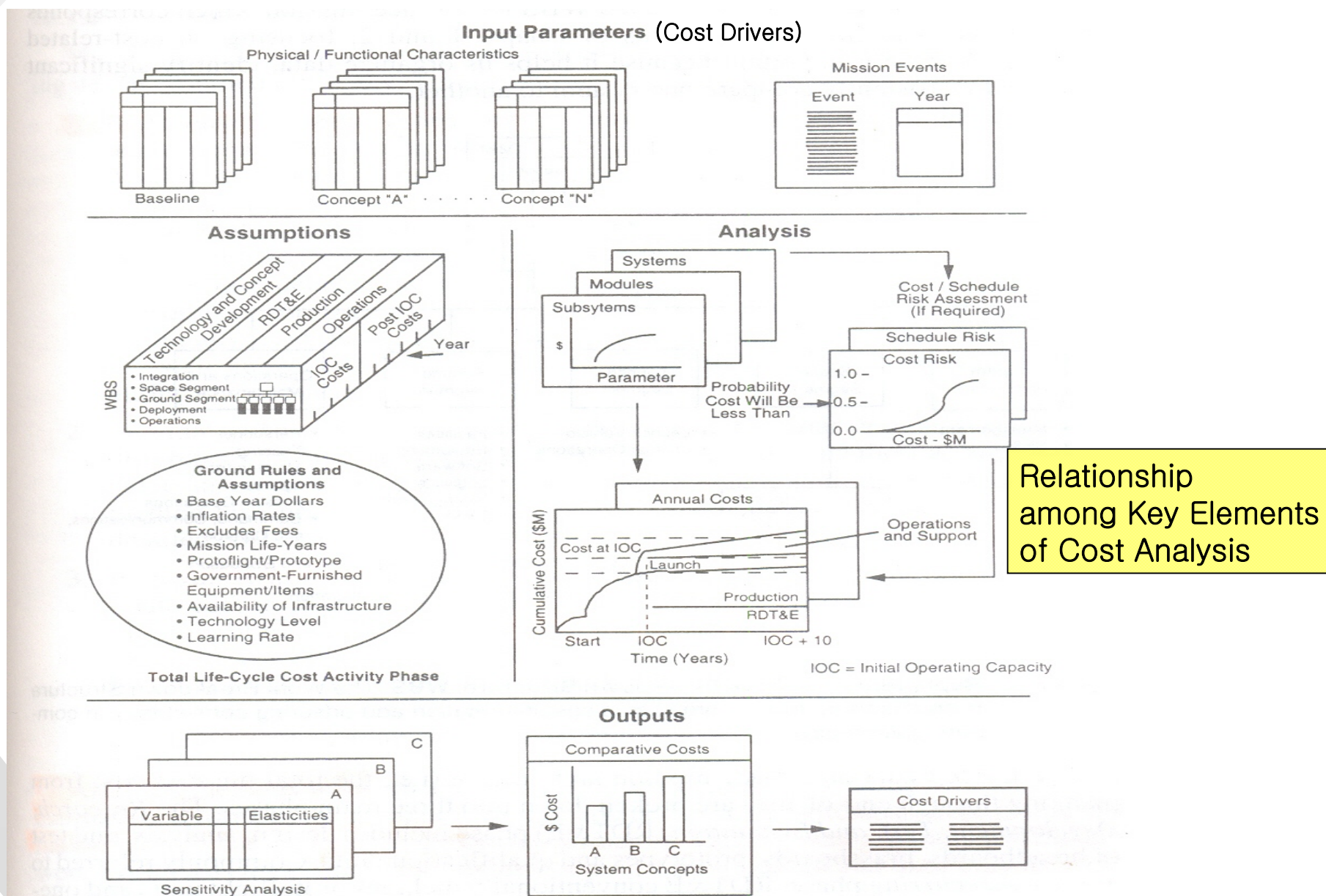


Fig. 20-1. Life-Cycle Cost Analysis Approach. This analysis approach is comprehensive and provides cost information necessary to properly evaluate system candidates.

20.1 Intro. to Cost Analysis

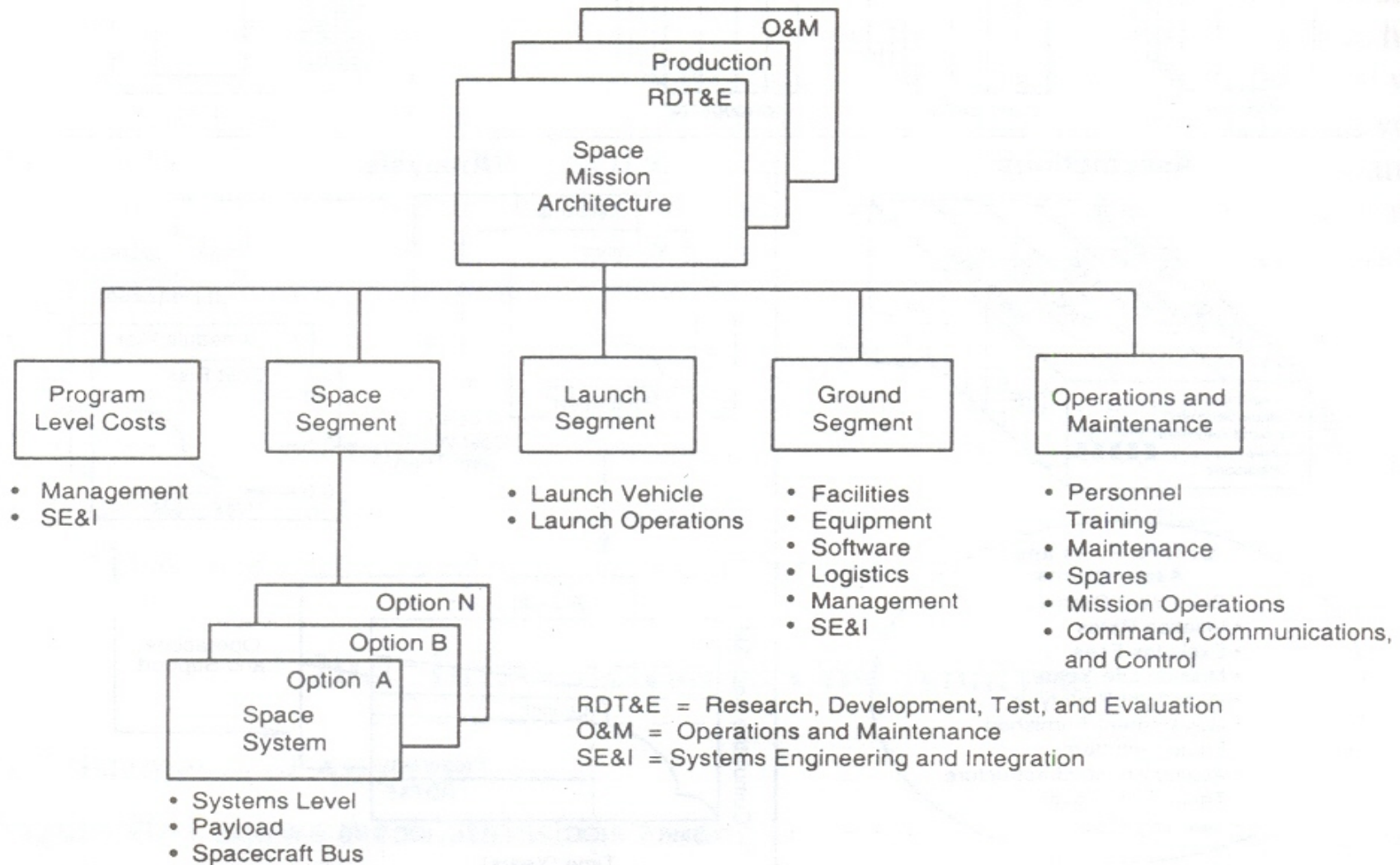
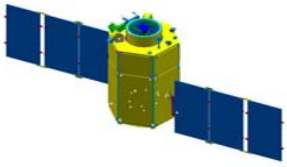


Fig. 20-2. Representative Work Breakdown Structure, WBS. The Work Breakdown Structure is an important tool for organizing cost information and ensuring consistency in comparing alternatives.



20.1 Intro. to Cost Analysis

◆ Cost Estimation Method

■ Three Basic Methods

■ Detailed Bottom-Up Estimation

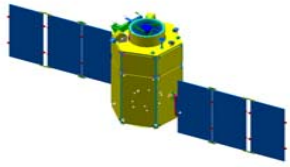
- Identify and specify at a low level elements that make up the system
- Then estimate the cost of materials and labor to develop and produce each element

■ Analogy-Base Estimation

- Use the cost of a similar item and adjust it for differences in size or complexity

■ Parametric Estimation/Cost Estimation Relationship (CER) equation

- Use a series of mathematical relationship that relate cost to physical, technical, and performance parameters
- Three General Sources for Parametric Cost Estimation Models
 - Publicly available special purpose models
 - Publicly available general purpose models
 - Private specific purpose models



20.1 Intro. to Cost Analysis

◆ Cost Model Development

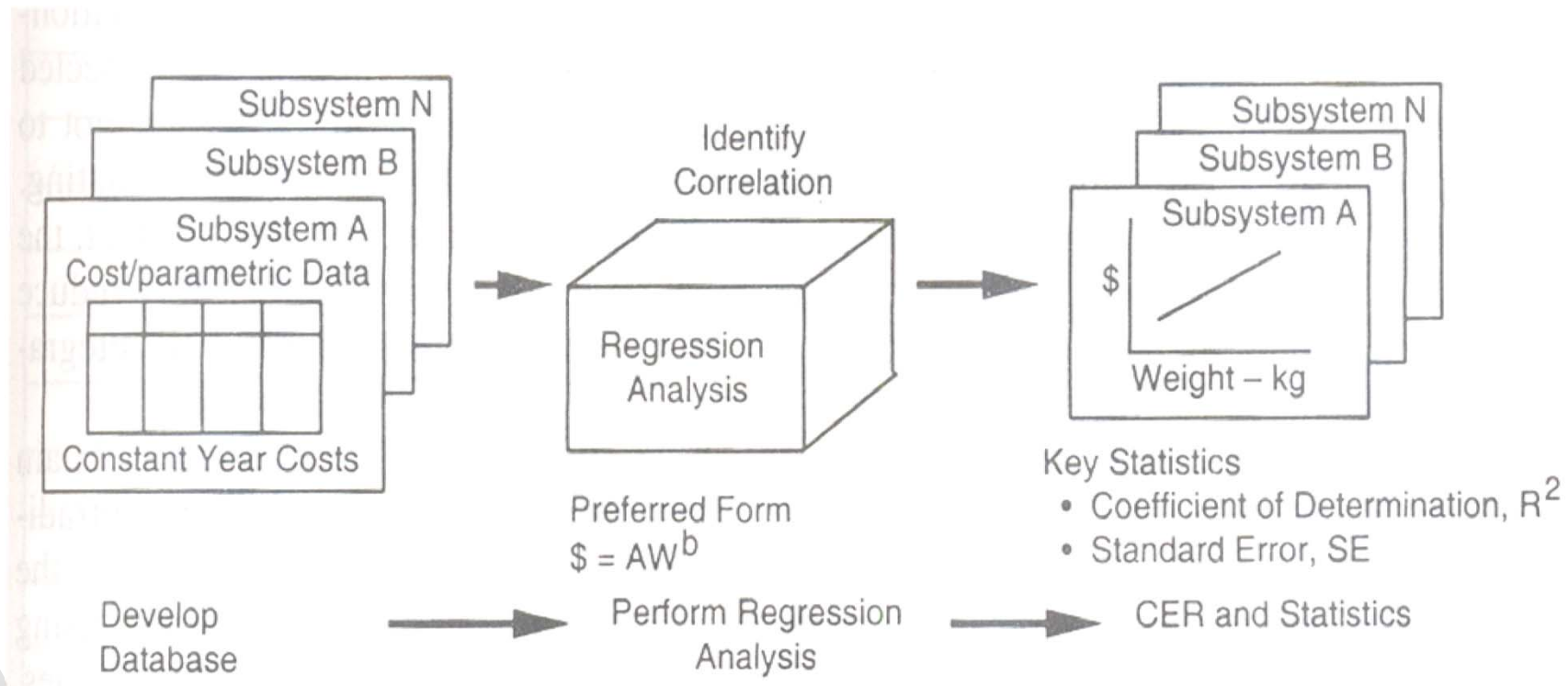
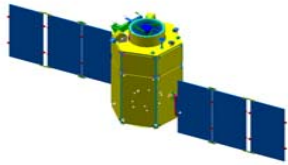


Fig. 20-3. Deriving CERs from Actuals. The key to a credible CER is an accurate database of relevant historical information from actual programs.

- After cost data are properly categorized and normalized, the task of CER development begins

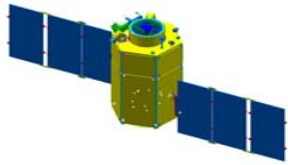


20.1 Intro. to Cost Analysis

◆ Types of Dollars

TABLE 20-1. Inflation Factors Relative to the Year 2000 Based on Projections by the Office of the Secretary of Defense (January 1998). See text for discussion.

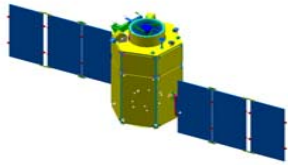
Fiscal Year (FY)	Inflation Factor to Base Year 2000	Fiscal Year (FY)	Inflation Factor to Base Year 2000
1980	0.456	2001	1.017
1981	0.510	2002	1.034
1982	0.559	2003	1.052
1983	0.610	2004	1.075
1984	0.658	2005	1.099
1985	0.681	2006	1.123
1986	0.700	2007	1.148
1987	0.719	2008	1.173
1988	0.740	2009	1.199
1989	0.771	2010	1.225
1990	0.802	2011	1.252
1991	0.837	2012	1.279
1992	0.860	2013	1.308
1993	0.883	2014	1.336
1994	0.901	2015	1.366
1995	0.918	2016	1.396
1996	0.937	2017	1.427
1997	0.958	2018	1.458
1998	0.970	2019	1.490
1999	0.984	2020	1.523
2000	1.000		



20.2 The Parametric Cost Estimation Process

TABLE 20-2. Parametric Cost Estimation Process. This process provides an estimate of the total life-cycle cost.

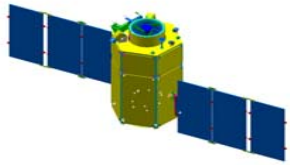
Step	Reference
1. Develop Work Breakdown Structure – Identify all cost elements	Fig. 20-2
2. List Space System Characteristics – Identify advanced technology parameters	Table 20-3 Table 20-8
3. Compute Space Segment Cost – RDT&E cost – Software cost – Theoretical first unit cost – Subsequent unit costs	Tables 20-4, 20-6, 20-9 Table 20-10 Tables 20-5, 20-6, 20-9 Sec. 20.4
4. Compute Launch Segment Cost	Table 20-14
5. Compute Ground Segment Cost – First ground station – Software cost – Additional ground stations – Earth terminals	Table 20-11 Table 20-10 Table 20-11 Table 20-13
6. Compute Operations and Maintenance Cost – Space segment spares – Launch costs for spares – Ground system operations and support	Sec. 20.5 Table 20-14 Table 20-12
7. LIFE-CYCLE COST	Sum of items 3–6 above



20.2 Parametric Cost Estimation

TABLE 20-3. Space Mission Characteristics Required for Parametric Cost Modeling.

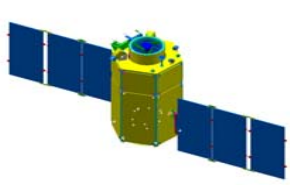
Characteristic	Reference	FireSat Example
1. Constellation		
No. of spacecraft in constellation	Secs. 7.1, 7.6	2
Orbit altitude	Secs. 3.3, 7.4	700 km
2. External Communications Resources		
TDRS	Secs. 13.1, 13.2	No
DomSat	Secs. 13.1, 13.2	No
3. Space Segment		
Payload		
Type	Sec. 9.1	IR
Weight (Communications)	Sec. 13.4	N/A
Aperture (IR, Visible)	Sec. 9.3	0.26 m
Spacecraft bus		
Dry weight by subsystem	Secs. 10.2, 10.3	Table 10-30
Volume	Sec. 10.5.1	1.7 m ³
Pointing accuracy	Sec. 11.1	0.1 deg
Pointing knowledge	Sec. 11.1	< 0.1 deg
Stabilization type	Sec. 10.4.2	3-axis
Flight software lines of code	Sec. 16.2	26 K
Average power	Sec. 10.6	110 W
BOL power	Sec. 10.6	NA
EOL Power	Sec. 10.4.2	NA
Solar array area	Sec. 10.6	8.5 m ²
Battery capacity	Sec. 11.4.2	17.5 A-hr
Data storage capacity	Sec. 11.3	
Number of thrusters	Sec. 11.1.4	4
Space segment design life	Sec. 10.4	5 yrs.



20.2 Parametric Cost Estimation

4. Launch Segment		
Launch vehicle	Secs. 18.1, 18.2	Pegasus
Upper stage	Secs. 18.1, 18.2	None
Launch site	Secs. 18.1, 18.2	N/A
No. of spacecraft per launch	Secs. 18.1, 18.2	1
5. Ground Segment		
No. of fixed and mobile sites	Secs. 13.1, 15.6	1 fixed
Software language	Sec. 16.3	Ada
Lines of code	Sec. 16.3	100 K
New or existing equipment and facilities	Sec. 15.2	New
Communications operating frequency	Sec. 13.3.5	S-band
6. Mission Operations and Support		
Mission duration from IOC*	Sec. 1.4	10 yrs.
No. of personnel	Sec. 14.3	10
No. of spare spacecraft	Sec. 19.1	0
No. of Shuttle support flights	Sec. 19.1	0

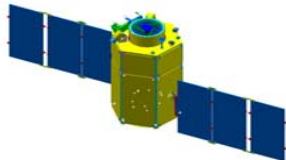
*IOC = Initial Operating Capability = time of launch of first satellite to provide operational data.



20.3 Cost Estimation Relationship

◆ Space Segment Cost

- Cost Estimation Relationship (CER)
 - Presented for preliminary mission analysis and design
 - Derived using different satellite data, statistical framework, and error models as noted
 - Some CERs provide total subsystem cost while others estimate RDT&E and TFU (Production Phase) separately
- Wraps
 - Cost associated with labor-intensive activities where a level of manpower is allocated over some period of performance
- Non-Recurring Costs
 - Include all costs with design verification and interface requirements
- Recurring Costs
 - Cover all efforts associated with flight hardware manufacture, IA&T (Integration, Assembly & Test), and a portion of program management and system engineering costs
- Heritage
 - defined as the percentage of a subsystem that is identical to previous S/C



20.3 CER

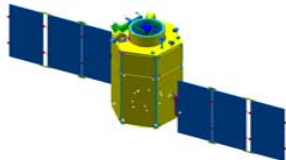
TABLE 20-4. CERs for Estimating Subsystem RDT&E Cost (FY00\$K). Applicable range for a good estimate is 25% above and below this data range. CER represents contractor cost without fee.

Cost Component	Parameter, X (Unit)	Input Data Range	RDT&E CER* (FY00\$K)	SE (%)
1. Payload				
1.1 IR Sensor	aperture dia. (m)	0.2–1.2	$356,851 X^{0.562}$	53,559 [†]
1.2 Visible Light Sensor	aperture dia. (m)	0.2–1.2	$128,827 X^{0.562}$	19,336 [‡]
1.3 Communications	comm. subsystem wt. (kg)	65–395	$353.3 X$	51
2. Spacecraft	spacecraft dry wt. (kg)	235–1,153	$101 X$	33
2.1 Structure	structure wt. (kg)	54–392	$157 X^{0.83}$	38
2.2 Thermal	X_1 = thermal wt. (kg)	3–48	$394 X_1^{0.635}$	45
	X_2 = spacecraft wt. + payload wt. (kg)	210–404	$1.1 X_1^{0.610} X_2^{0.943}$	32
2.3 Electrical Power System (EPS)	X_1 = EPS wt. (kg)	31–491	$62.7 X_1$	57
	X_2 = BOL power (W)	100–2,400	$2.63 (X_1 X_2)^{0.712}$	36
2.4 Telemetry, Tracking & Command (TT&C)/DH [‡]	TT&C/DH wt. (kg)	12–65	$545 X^{0.761}$	57
2.5 Attitude Determination & Control Sys. (ADCS)	ADCS wt. (kg)	20–160	$464 X^{0.867}$	48
2.6 Apogee Kick Motor (AKM)	AKM wt. (kg)	81–966	$17.8 X^{0.75}$	—
3. Integration, Assembly & Test (IA&T)	spacecraft bus + payload total RDT&E cost (FY00\$K)	2,703 – 395,529	$989 + 0.215 X$	46
4. Program Level	spacecraft bus + payload total RDT&E cost (FY00\$K)	4,607 – 523,757	$1.963 X^{0.841}$	36
5. Ground Support Equipment (GSE)	spacecraft bus + payload total RDT&E cost (FY00\$K)	24,465 – 581,637	$9.262 X^{0.642}$	34
6. Launch & Orbital Operations Support (LOOS)	N/A			

* Taken from USCM, 7th edition (1994) using minimum, unbiased percentage error CERs.

[†] Absolute error (in FY00\$K), not percentage error.

[‡] Includes spacecraft computer. If separate CERs for TT&C and C&DH are desired, use a 0.45/0.55 split



20.3 CER

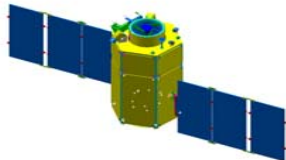
TABLE 20-5. CERs for Estimating Subsystem Theoretical First Unit (TFU) Cost.

Cost Component	Parameter, X (Unit)	Input Data Range	TFU CER* (FY00\$K)	SE (%)
1. Payload				
1.1 IR Sensor	aperture dia. (m)	0.2–1.2	$142,742 X^{0.562}$	21,424†
1.2 Visible Light Sensor	aperture dia. (m)	0.2–1.2	$51,469 X^{0.562}$	7,734†
1.3 Communications	comm. subsystem wt. (kg)	65–395	140 X	43
2. Spacecraft	spacecraft dry wt. (kg)	154–1,389	43 X	36
2.1 Structure	structure wt. (kg)	54–560	13.1 X	39
2.2 Thermal	thermal wt. (kg)	3–87	$50.6 X^{0.707}$	61
2.3 Electrical Power System (EPS)	EPS wt. (kg)	31–573	$112 X^{0.763}$	44
2.4 Telemetry, Tracking & Command (TT&C)/DH‡	TT&C/DH wt. (kg)	13–79	$635 X^{0.568}$	41
2.5 Attitude Determination & Control Sys. (ADCS)	ADCS wt. (kg)	20–192	$293 X^{0.777}$	34
2.6 Apogee Kick Motor (AKM)	AKM wt. (kg)	81–966	$4.97 X^{0.823}$	20
3. Integration, Assembly & Test (IA&T)	spacecraft bus wt. payload wt. (kg)	155–1,390	10.4 X	44
4. Program Level	spacecraft + payload total recurring cost (FY00\$K)	15,929 – 1,148,084	0.341 X	39
5. Ground Support Equipment (GSE)	N/A			
6. Launch & Orbital Operations Support (LOOS)	spacecraft bus + payload wt. (kg)	348–1,537	4.9 X	42

* Taken from USCM, 7th edition (1994) using minimum, unbiased percentage error CERs.

† Absolute error (FY00\$K), not percentage error.

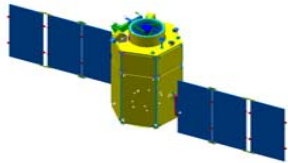
‡ Includes spacecraft computer. If separate CERs for TT&C and C&DH are desired, use a 0.45/0.55 split.



20.3 CER

TABLE 20-6. Cost-Estimating Relationships for Earth-orbiting Small Satellites Including RDT&E and Theoretical First Unit. Total subsystem cost in FY00\$M as a function of the independent variable, X.

Cost Component	Parameter, X (Unit)	Input Data Range	Subsystem Cost CER* (FY00\$K)	SE (FY00\$K)
1. Payload	Spacecraft Total Cost (FY00\$K)	1,922–50,651	0.4 X	$0.4 \times SE_{bus}$
2. Spacecraft	Satellite bus dry wt. (kg)	20–400	$781 + 26.1 X^{1.261}$	3,696
2.1 Structure†	Structures wt. (kg)	5–100	$299 + 14.2 X \ln(X)$	1,097
2.2 Thermal‡	Thermal control wt. (kg)	5–12	$246 + 4.2 X^2$	119
	Average power (W)	5–410	$-183 + 181 X^{0.22}$	127
2.3 Electrical Power System (EPS)	Power system wt. (kg)	7–70	$-926 + 396 X^{0.72}$	910
	Solar array area (m ²)	0.3–11	$-210,631 + 213,527 X^{0.0066}$	1,647
	Battery capacity (A-hr)	5–32	$375 + 494 X^{0.754}$	1,554
	BOL Power (W)	20–480	$-5,850 + 4,629 X^{0.15}$	1,585
	EOL Power (W)	5–440	$131 + 401 X^{0.452}$	1,603
2.4a Telemetry Tracking & Command (TT&C)**	TT&C/DH wt. (kg)	3–30	$357 + 40.6 X^{1.35}$	629
	Downlink data rate (Kbps)	1–1,000	$3,636 - 3,057 X^{-0.23}$	1,246
2.4b Command & Data Handling (C&DH)	TT&C + DH wt. (kg)	3–30	$484 + 55 X^{1.35}$	854
	Data Storage Capacity (MB)	0.02–100	$-27,235 + 29,388 X^{0.0079}$	1,606



20.3 CER

2.5 Attitude Determination & Control Sys. (ADCS)	ADCS dry wt. (kg)	1–25	$1,358 + 8.58 X^2$	1,113
	Pointing accuracy (deg)	0.25–12	$341 + 2651 X^{-0.5}$	1,505
	Pointing knowledge (deg)	0.1–3	$2,643 - 1,364 \ln(X)$	1,795
2.6 Propulsion††	Satellite Bus dry wt. (kg)	20–400	$65.6 + 2.19 X^{1.261}$	310
	Satellite volume (m ³)	0.03–1.3	$1539 + 434 \ln(X)$	398
	Number of Thrusters	1–8	$4,303 - 3,903 X^{-0.5}$	834
3. Integration, Assembly & Test (IA&T)	Spacecraft total cost (FY00\$K)	1,922 – 50,651‡‡	0.139 X	$0.139 \times SE_{bus}$
4. Program Level	Spacecraft total cost (FY00\$K)	1,922 – 50,651‡‡	0.229 X	$0.229 \times SE_{bus}$
5. Ground Support Equipment (GSE)	Spacecraft total cost (FY00\$K)	1,922 – 50,651‡‡	0.066 X	$0.066 \times SE_{bus}$
6. Launch & Orbital Operations Support (LOOS)	Spacecraft total cost (FY00\$K)	1,922 – 50,651‡‡	0.061 X	$0.061 \times SE_{bus}$

* CERs based on the **Small Satellite** Cost Model [Bearden, Boudreault, and Wertz, 1996], adjusted for inflation as shown in Table 20-1, and broken into subsystem cost using the percentages from Table 20-9.

† Aluminum materials primarily with selected use of advanced materials (e.g. composites, magnesium).

‡ Thermal CER appropriate for passive systems only.

** CER applies to UHF/VHF and S-band LEO systems

†† Hydrazine monopropellant and cold-gas stationkeeping systems only. CER not appropriate for bipropellant or dual-mode systems. Costs of AKM are not included.

‡‡ Input data range for items 3–6 calculated using min and max values of input data range for spacecraft bus cost CER in item 2.

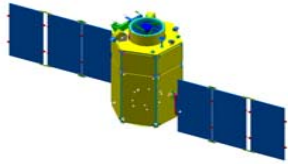


TABLE 20-7. Allocation of Program-Level Cost. We allocate the program level or wrap costs in Tables 20-4, 20-5, and 20-6 to their components as shown.

Program Level Component	RDT&E	Theoretical First Unit
Program Management	20%	30%
Systems Engineering	40%	20%
Product Assurance	20%	30%
System Evaluation	20%	20%
Total	100%	100%

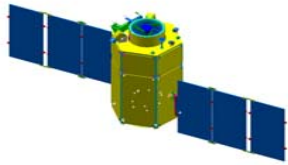
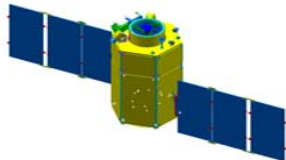


TABLE 20-8. Heritage Cost Factors. We apply these factors to the CERs and their standard errors from Tables 20-4, 20-5, and 20-6 as described in the text. Data from Hamaker [1987].

Multiplicative Factors for Development Heritage (Apply to RDT&E Costs Only)	
New design with advanced development	> 1.1
Nominal new design—some heritage	1.0
Major modification to existing design	0.7 – 0.9
Moderate modifications	0.4 – 0.6
Basically existing design	0.1 – 0.3



20.3 CER

TABLE 20-9. Breakdown of Small Satellite Costs. These factors can be applied to overall small spacecraft costs to estimate the cost of constituent subsystems and the RDT&E vs. TFU costs. In the first column, 100%; Total spacecraft hardware cost, excluding payload and wraps. Data on cost breakdown by subsystem courtesy Microcosm; derived from Wertz and Larson [1996] and Sarsfield [1998]. Comparable factors can be found in Bearden [1999] and Sadin and Davis [1993]. Recurring vs. Nonrecurring estimates are from Bearden, Burgess, and Lao [1995]. The final row represents the total system cost exclusive of launch cost and operations.

Subsystem/Activity	Fraction of Spacecraft Bus Cost (%)	Non-Recurring Percentage (%)	Recurring Percentage (%)
1.0 Payload	40.0%	60%	40%
Bus Total	100.0%	60%	40%
2.1 Structure	18.3%	70%	30%
2.2 Thermal	2.0%	50%	50%
2.3 EPS	23.3%	62%	38%
2.4a TT&C	12.6%	71%	29%
2.4b C&DH	17.1%	71%	29%
2.5 ADCS	18.4%	37%	63%
2.6 Propulsion*	8.4%	50%	50%
Wraps			
3.0 IA&T (Integration, Assembly & Test)	13.9%	0%	100%
4.0 Program Level (Contractor cost)	22.9%	50%	50%
5.0 GSE (Ground Support Equipment)	6.6%	100%	0%
6.0 LOOS (Launch and Orbital Operations Support)	6.1%	0%	100%
Total	189.5%	92.0%	97.5%

* Propulsion costs may be excluded if, as is the case with many small satellites, the spacecraft doesn't have a propulsion system.

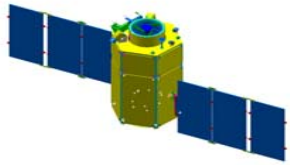


TABLE 20-10. Software Development Costs. RDT&E costs only (in FY00\$K). See Sec. 16.2.2 for estimates of the lines of code.

Flight Software	435 × KLOC
Ground Software	220 × KLOC
KLOC = Thousand of Lines of Code; cost without fee	
FACTORS FOR OTHER LANGUAGES	
Language	Factor
Ada	1.00
UNIX-C	1.67
PASCAL	1.25
FORTTRAN	0.91

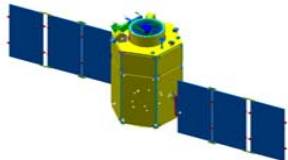


TABLE 20-11. Ground Segment Development Cost Model. For preliminary mission design this should be used in conjunction with the software cost estimate from Table 20-10.

Ground Station Element	Development Cost Cost Distribution (%)	Development Cost as Percent of Software Cost (%)
Facilities (FAC)	6	18
Equipment (EQ)	27	81
Software (SW)	33	100(Reference)
Logistics	5	15
Systems Level		
Management	6	18
Systems Engineering	10	30
Product Assurance	5	15
Integration and Test	8	24
Total	100	

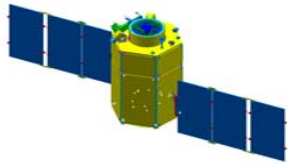
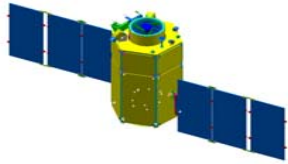


TABLE 20-12. Operations and Support Cost in FY00\$. See text for details.

Maintenance	$0.1 \times (\text{SW} + \text{EQ} + \text{FAC})/\text{year}$
Contractor Labor	\$160K/Staff Year
Government Labor	\$110K/Staff Year

TABLE 20-13. Earth Terminals, Antennas, and Communication Electronics. Costs are for hardware only, and assume attachment to existing facilities [DCA, 1996].

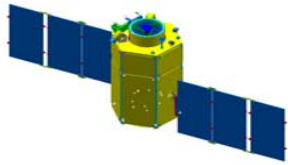
Frequency	Cost (FY00\$K)
SHF	$(50 \times D) + (400 \times P) + 1,800$
K, C Band	640
Ku Band	750
D = antenna diameter in m	P = RF power in kW



20.3 CER

TABLE 20-14. Launch Vehicle Costs in FY00\$M. The data assumes launch from the country's main site. Except where noted, LEO altitude is 185 km and inclination is 28.5 deg (5.2 deg for Ariane). Data from Isakowitz [1995].

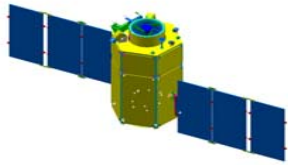
Launch Vehicles	Maximum Payload-to-Orbit (kg)			Unit Cost (FY00\$M)	Cost per kg to LEO (FY00\$K/kg)
	LEO	GTO	GEO		
USA					
Atlas II	6,580	2,810		80–90	12.2—13.7
Atlas II A	7,280	3,039		85–95	11.7–13.0
Atlas II AS	8,640	3,606		100–110	11.6–12.7
Athena 1	800			18	22.5
Athena 2	1,950			26	13.3
Athena 3	3,650			31	8.5
Delta II (7920, 7925)	5,089	1,840		50–55	9.8–10.8
Pegasus XL	460			13	28.3
Saturn V	127,000			820	6.5
Shuttle* (IUS or TOS)	24,400	5,900	2,360	400	16.4
Titan II	1,905			37	19.4
Titan IV	21,640	8,620	5,760 (Centaur)	214 (270)	9.9
Taurus	1,400	450		20–22	14.3–15.7



20.3 CER

ESA					
Ariane 4 (AR40)	4,900	2,050		50–65	10.2–13.3
Ariane 4 (AR42P)	6,100	2,840		65–80	10.7–13.1
Ariane 4 (AR44L)	9,600	4,520		95–120	9.9–12.5
Ariane 5 (550 km)	18,000	6,800		130	7.2
CHINA					
Long March C23B	13,600	4,500	2,250	75	5.5
RUSSIA					
Proton SL-13	20,900			55–75	2.6–3.6
Kosmos C-1	1,400			11	7.9
Soyuz	7,000			13–27	1.9–3.9
Tsyklon	3,600			11–16	3.1–4.4
Zenit 2	13,740			38–50	2.8–3.6
JAPAN					
H-2	10,500	4,000	2,200	160–205	15.2–19.5
J-1	900			55–60	61.1–66.7
GTO = Geosynchronous Transfer Orbit; GEO = Geostationary Orbit; LEO = Low-Earth Orbit					

* There is no official price for a Space Shuttle launch. Following the Challenger loss, only government payloads have been allowed. The GAO has assigned a price of \$400 million per flight, but the actual cost depends strongly on the flight rate.

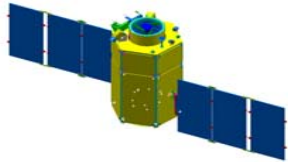


20.4 Other Topics

TABLE 20-15. Technology Classification and Relative Cost Risk. Definitions are from NASA.

Technology Readiness Level	Definition	Relative Risk Level	Standard Deviation about MLE (%)
1	Basic principles observed	High	> 25
2	Conceptual design formulated	High	> 25
3	Conceptual design tested analytically or experimentally	Moderate	20–25
4	Critical function/characteristic demonstrated	Moderate	15–20
5	Component or breadboard tested in relevant environment	Moderate	10–15
6	Prototype/engineering model tested in relevant environment	Low	< 10
7	Engineering model tested in space	Low	< 10
8	Full operational capability	Low	< 10

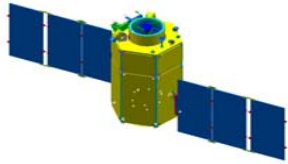
Most Likely Estimate



20.4 Other Topics

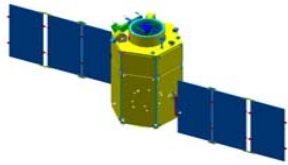
TABLE 20-16. Space Systems Costs. This table can be used to obtain quick order-of-magnitude estimates. All values are in FY00\$M.

Manned Space Programs		Total Program Costs(\$M)
Apollo		152,000
Orbiter		45,000
Gemini		4,400
SkyLab		3,100
Mercury		1,100
Space Observatories		Total Program Costs
Space Telescope		2,270
GRO		640
HETE		31
Sampex		75



20.4 Other Topics

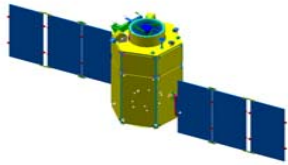
Communication Satellites	Dry Weight (kg)	Average Unit Costs (\$M)	\$K/kg
Intelsat VIII (commercial)	1,200	133	111
TDRSS (NASA)	1,550	126	81
DSCS IIIB (DoD)	806	114	141
Westar (commercial)	500	78	156
ORBCOMM	33	11	333
Surveillance/Navigation Satellites			
DSP	2,200	314	143
GPS-2	839	57	68
Meteorological Satellites			
GOES	500	84	168
DMSP	514	88	171
Interplanetary Spacecraft			
Pioneer (S/C bus only)	231	38	165
Mars Observer	1,018	77	76
Clementine	232	57	246
Experimental Small Satellites			
RADCAL	92	5	54
PoSAT-1	49	1.2	24
AMSAT AO-13	84	1	12
Freja	214	19	89
Ørsted	60	15	243



20.5 FireSat Example

TABLE 20-17. FireSat Space and Launch Segment Costs in FY00\$K. Cost estimates based on data in Table 20-3 Tables 20-4 and 20-5 for traditional satellite designs. Input data from Tables 10-31 and 20-3.

Cost Component	Parameter, Value, Data Source	RDT&E Cost (FY00\$K)	1st Unit Cost (FY00\$K)	2nd Unit Cost (FY00\$K)	Total Cost (FY00\$K)	SE (\$K)
1. Payload						
1.1 IR Sensor	aperture dia. = 0.263 m	168,462	67,386	60,647	296,495	94,265
2. Spacecraft bus						
2.1 Structure*	wt. = 32.0 kg	2,784	419	377	3,580	1,368
2.2 Thermal*	wt. = 6.8 kg	1,337	197	177	1,712	830
2.3 Electrical Power System (EPS)*	wt. = 45.7 kg	2,862	2,067	1,860	6,790	3,360
2.4 Telemetry Tracking & Command and Data Handling (TT&C/DH)*	wt. = 6.8 kg	2,356	1,894	1,705	5,955	2,819
2.5 Attitude Determination & Control Sys. (ADCS)*	wt. = 18.3 kg	5,753	2,799	2,519	11,071	4,570
2.6 Propulsion	NA†					
Spacecraft Bus Total Cost		15,092	7,376	6,639	29,107	9,739



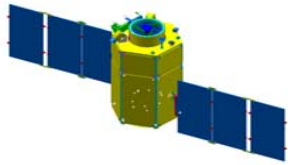
20.5 FireSat Example

3. Integration, Assembly & Test (IA&T)	Spacecraft bus + payload RDT&E cost = \$183,554	40,453	1,456	1,310	43,220	19,826
4. Program Level	Same as previous	52,450	25,494	22,944	100,888	37,773
5. Ground Support Equipment (GSE)	Same as previous	22,184	—	—	22,184	7,543
6. Launch & Orbital Ops Support (LOOS)	Spacecraft + payload wt. = 140 kg	—	686	617	1,303	547
7. Flight Software	26 KLOC	5,655†	0	0	5,655	—
Total Space Segment Cost to Contractor		304,297	102,398	92,158	498,853	
10% Contractor Fee		30,430	10,240	9,216	49,885	
Total Space Segment Cost to Government		334,727	112,638	101,374	548,738	
8. Launch Segment	2 Pegasus XL Launchers		13,000	13,000	26,000	
Total Cost of Deployment					574,738	136,947

* Spacecraft bus subsystem masses shown include a fraction of the spacecraft mass margin of 11.2 kg.

† FireSat propulsion system is taken into account in ADCS CER (2.5). The 18.3 kg value for ADCS mass includes propulsion system hardware mass.

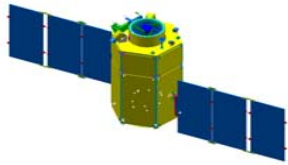
‡ Assumes a heritage cost factor of 0.5 (i.e., moderate modifications to existing flight software). Calculated using flight software CER in Table 20-10.



20.5 FireSat Example

TABLE 20-18. FireSat Space and Launch Segment Costs in FY00\$K. Based on data in Table 20-3, Small Satellite CERs in Table 20-6 and ratios for other cost components in Table 20-9.

Cost Component	Parameter, Value, Data Source	RDT&E Cost (FY00\$K)	1st Unit Cost (FY00\$K)	2nd Unit Cost (FY00\$K)	Total Cost (FY00\$K)	SE (\$K)
1. Payload	Spacecraft total cost (RDT&E + TFU)	3,049	2,033	1,829	6,911	1,946
2. Spacecraft bus						
2.1 Structure*	wt. = 32.0 kg	1,318	565	508	2,390	1,393
2.2 Thermal*	wt. = 6.8 kg	221	221	199	642	173
2.3 Electrical Power System (EPS)*	wt. = 45.7 kg	3,271	2,005	1,804	7,080	1,221
2.4a Telemetry Tracking & Command (TT&C)*	TT&C + DH wt. = 6.8 kg	641	262	235	1,138	793
2.4b Command & Data Handling (C&DH)*	TT&C + DH wt. = 6.8 kg	868	355	319	1,542	1,077
2.5 Attitude Determination & Control Sys. (ADCS)*	wt. = 9.1 kg,	767	1,306	1,175	3,247	1,744
2.6 Propulsion	Satellite bus dry mass = 112 kg	453	453	408	1,314	450
Spacecraft Bus Total Cost		7,538	5,166	4,649	17,353	4,883

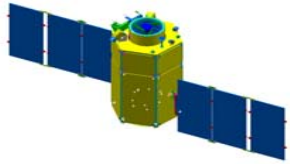


20.5 FireSat Example

3. Integration, Assembly & Test (IA&T)	Spacecraft total cost (RDT&E + TFU)	—	1,766	1,589	3,355	945
4. Program Level	Same as previous	1,455	1,455	1,309	4,218	1,188
5. Ground Support Equipment (GSE)	Same as previous	838	—	—	838	236
6. Launch & Orbital Ops Support (LOOS)	Same as previous	—	775	697	1,472	415
7. Flight Software	26 KLOC	5,655†	0	0	5,655	—
Total Space Segment Cost to Contractor		18,535	11,194	10,074	39,803	
10% Contractor Fee		1,854	1,119	1,007	3,980	
Total Space Segment Cost to Government		20,389	12,313	11,082	43,783	
8. Launch Segment	2 Pegasus XL Launchers		13,000	13,000	26,000	
Total Cost of Deployment					69,783	7,552

* Spacecraft bus subsystem masses shown include a fraction of the spacecraft mass margin of 11.2 kg.

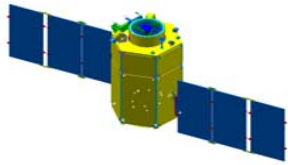
† Assumes a heritage cost factor of 0.5 (i.e., moderate modifications to existing flight software). Calculated using flight software CER in Table 20-10.



20.5 FireSat Example

TABLE 20-19. FireSat Annual Operations and Maintenance Cost in FY00\$M.

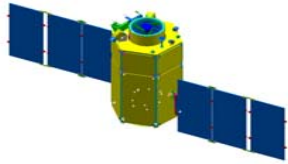
Operations and Maintenance	
10 Contractor Personnel (\$160K/Yr) including fee	1.6
Maintenance	4.4
Total Annual Cost	6



20.5 FireSat Example

TABLE 20-20. FireSat Ground Segment and Operations Costs in FY00\$M.

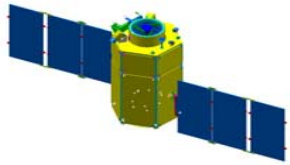
Development	Cost
Software 100 KLOC (Ada) @ \$220/LOC	22.0
Equipment	17.8
Facilities	<u>4.0</u>
Subtotal	43.8
Management	4.0
Systems Engineering	6.6
Product Assurance	3.3
Integration and Test	5.3
Logistics	<u>3.3</u>
Total	66.3



20.5 FireSat Example

TABLE 20-21. FireSat Life-Cycle Cost Estimate. All costs in FY00\$M, including fee. Launch segment estimate is based on two Pegasus launches.

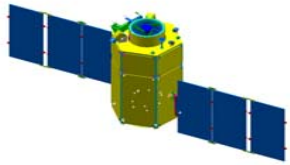
Initial Deployment	
Space Segment - Table 20-17	548.7
Launch Segment - Table 20-17	26.0
Ground Segment - Table 20-20	66.3
Subtotal	641.0
Operations and Maintenance - (Table 20-19)	
Annual Ops. and Maint.	6.0
Total Ops. and Maint. for 5 years	30.0
Total Life-Cycle Cost for 5 years	671.0



20.5 FireSat Example

TABLE 20-22. FireSat Cost Uncertainty

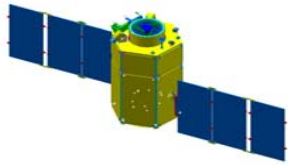
Element	TFU Cost (\$M)	System Technology Level	System Technology Std. Dev. (\$M)	Cost Estimate Std. Dev. (\$M)	Combined Std. Dev. (\$M)
S/C bus	7.4	6 (= 10%)	0.74	1.11	1.33
IR payload	67.4	5 (= 15%)	10.1	10.1	14.3
Total	74.8		Step 1: Sum		15.6
			Step 2: RSS		14.4
			Step 3: Average		15.0



20.5 FireSat Example

TABLE 20-23. Time Distribution of FireSat Costs. Based on a total recurring cost for the first two units of \$194.5M (Table 20-17) and assuming 60% of costs spent in the first 2.5 years.

Year	Cumulative Cost		Annual Cost	
	(%)	(\$M)	(%)	(\$M)
1	12.3	23.9	12.3	23.9
2	42.8	83.2	30.5	59.3
3	75.6	147	32.8	63.8
4	95.8	186.3	20.2	39.3
5	100.0	194.5	4.2	8.2
Totals			100.0	194.5



20.5 FireSat Example

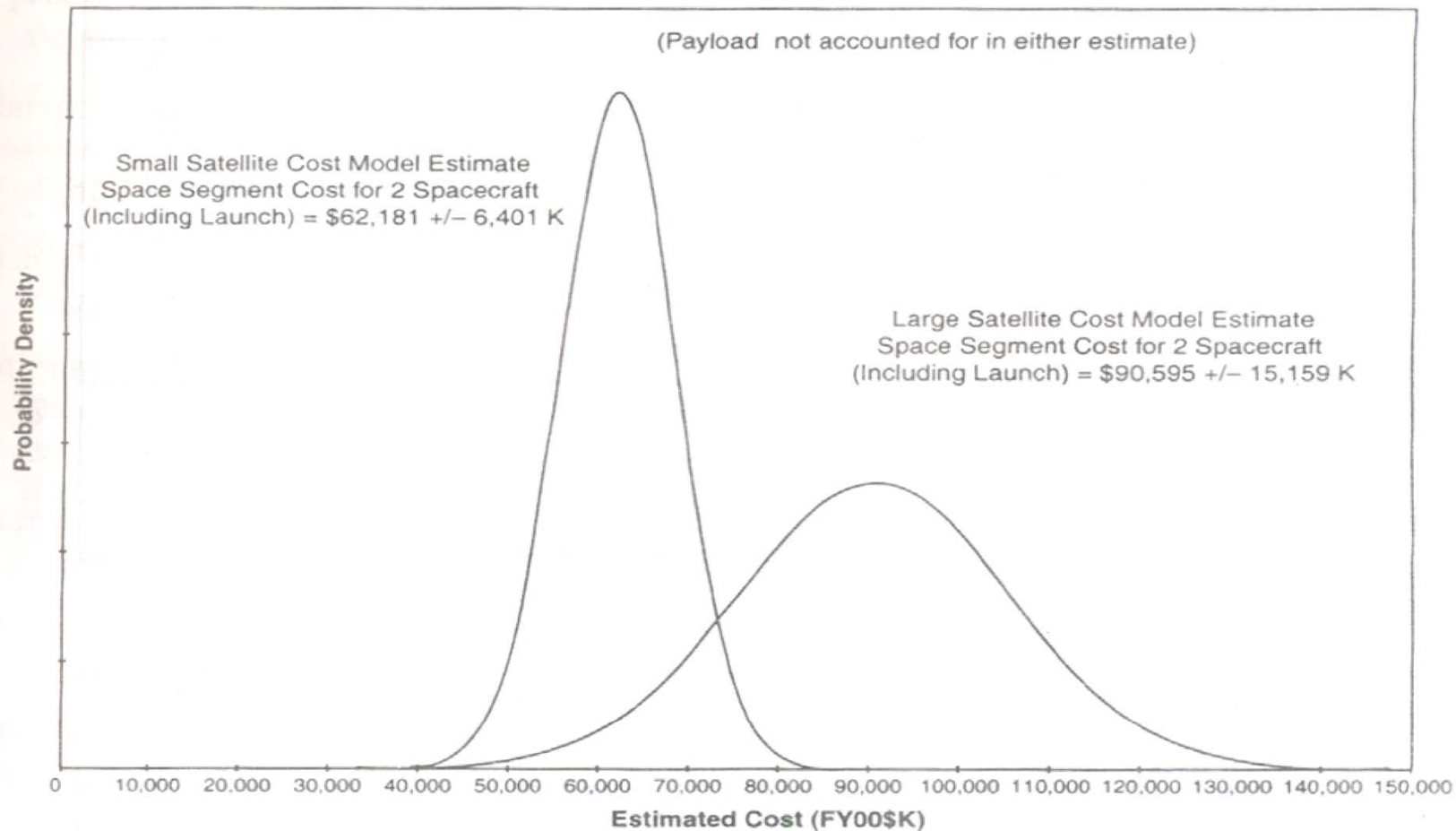
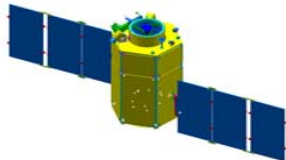


Fig. 20-4. Probability Density of Estimated Cost Resulting from Cost Risk Analysis. The taller, narrower curve to the left is the result for the small satellite CERs from Table 20-6 and the wider, lower curve is the result for the large satellite CERs from Tables 20-4 and 20-5. Most missions of similar scope to FireSat will lie somewhere between these two extremes. The results of such statistical evaluations can be used to dictate reserves policy (e.g., “risk” dollars) or to guide technology investment.



20.5 FireSat Example

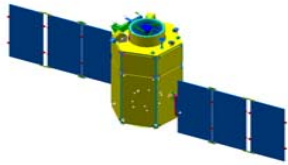
TABLE 20-24. FireSat Mass, Power and Cost as a Function of Design Life. Data assumes fixed performance requirements. Redundancy and associated impacts to subsystem masses, and increased power and propellant requirements drive the results.

Design Life (years)	1	3	5*	7
<i>Payload (kg)</i>	20	24	30	35
<i>Propulsion (kg)</i>	7	8	9	18
<i>ADCS (kg)</i>	7	9	9	22
<i>TT&C/C&DH (kg)</i>	5	6	7	20
<i>Thermal (kg)</i>	5	6	7	9
<i>Power (kg)</i>	33	40	46	70
<i>Structures (kg)</i>	22	27	32	37
<i>Spacecraft Dry Mass (kg)</i>	99	120	140	211
<i>Propellant (kg)</i>	31	33	35	37
<i>Launch Mass (kg)</i>	130	153	175	248
<i>Performance Pegasus (kg)</i>	290	290	290	290
<i>Launch Margin†</i>	55%	47%	40%	14%
<i>EOL Power (W)</i>	140	140	140	140
<i>BOL Power (W)</i>	145	157	170	183
<i>Space Segment Cost (FY00\$M)</i>	35	40	44	76
<i>Launch Cost (FY00\$M)</i>	26	26	26	36‡
<i>Operations Cost (FY00\$M)</i>	6	18	30	45
<i>Total Cost (FY00\$M)</i>	67	84	100	157
<i>Cost per Year (FY00\$M)</i>	67	28	20	22

* FireSat baseline design.

†25% margin required at conceptual design stage

‡Launch mass plus 25% margin exceeds payload capability of Pegasus XL. Two Athena 1 launches assumed at \$18M per launch.



20.5 FireSat Example

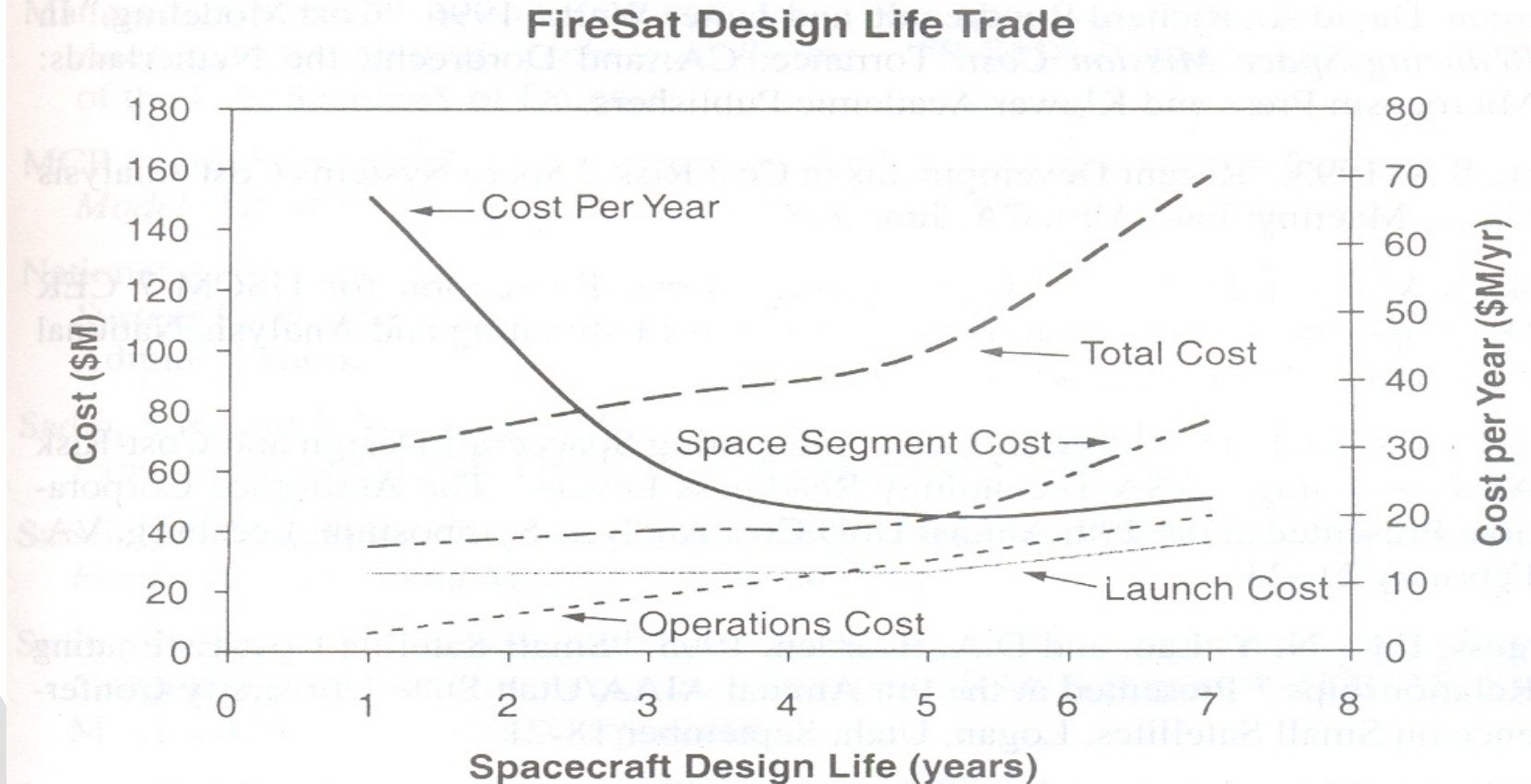


Fig. 20-5. FireSat Design Life Trade. Compare with Fig. 1-5 in Sec. 1.4. Total cost, the sum of individual spacecraft cost, launch vehicle cost, and operations cost, is shown as a function of spacecraft design life. On the right axis we plot cost per year which asymptotically approaches the yearly operations cost as design life goes to infinity. Optimal design life is chosen at a point on the cost-per-year curve that is relatively insensitive to changes in design life and while mission data still remains useful.